

Mohammad Ryyan Rashid

Research Software / Backend Engineer | Autonomous Systems & Applied AI

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Professional Summary

Research software and backend engineer in Abu Dhabi developing **TII-sponsored Unmanned Traffic Management systems** at Khalifa University, with prior machine-learning work as a **Research Intern at Harvard Medical School** and end-to-end ownership of a live financial-operations platform. Experience spans C++20 systems, Python/FastAPI services, full-stack delivery, computer vision, predictive modelling, and governed LLM integration.

Core Skills

Software engineering: C++20, CMake, Python, FastAPI, JavaScript, React/Next.js, Node.js, REST APIs

Systems and data: Concurrent processing, event-driven systems, NATS JetStream, SQL, PostgreSQL, MongoDB

AI and ML: PyTorch, TensorFlow, scikit-learn, computer vision, neural networks, NumPy, Pandas, LLM API integration

Delivery and operations: Docker, Git, Hostinger, DigitalOcean, GitHub Actions, CI/CD, Linux, systemd, backups

Professional Experience

Khalifa University

Research Assistant

Visiting Researcher

Abu Dhabi, UAE

Sep 2025 – Present

Jun 2025 – Sep 2025

- Develop and advance **PRISM**, a FastAPI-based web UTM simulator with an interactive frontend, configurable risk models, sequential/parallel execution, and timing/performance analysis, as part of **Technology Innovation Institute (TII)-sponsored deliverables**.
- Architect and develop a modular **C++20 real-time UAS conformance-monitoring system** that integrates research outputs, correlates tracking and Remote ID evidence, and evaluates identity, authorization, trajectory, and 3D airspace constraints through testable CMake modules and event-driven boundaries.
- Lead development of **Taqyeem**, a full-stack platform for outcome-based education and quality assurance with CLO–PLO mapping and interactive assessment workflows.

Aamina Charitable Trust

Pro Bono Software Engineer (15–20 hours/week)

Remote / Kashmir, India

Jun 2024 – Present

- Solely designed, built, and deployed a production platform, and continue to maintain it for a 10–12-person team managing **450+ beneficiary records across 370+ households, 4,600+ transactions**, and over **INR 1 million in monthly payment activity**.
- Reduced recurring financial administration from three people spending multiple hours per day to **less than 30 minutes total per day** through bank-statement processing, reconciliation, recurring payments, and digital approvals.
- Engineered permissions, audit trails, digital signatures, QR-backed verification, background jobs, authentication, backups, and a Telegram bot; built a governed OpenRouter interface with no direct database access.
- Own PostgreSQL, Docker, Hostinger deployment, Linux, and GitHub Actions CI/CD operations; the web system has operated since April 2026 with no significant outages.

Harvard Medical School

Research Intern

Boston, MA, USA

Apr 2023 – Oct 2023

- Developed and evaluated SVR, XGBR, MLR, and neural-network approaches for predicting hydrogel porosity, pore-size distribution, and mechanical properties.
- Applied image processing and genetic algorithms to biomaterials problems; related research contributed to a manuscript under peer review.

Selected Research and Publications

Visual Verification of UAV Location in Remote Identification Messages

IEEE / ICUAS 2026

Co-authored work on visually cross-verifying location information reported through UAV Remote ID messages.

IntelliBin: AI-Driven Waste Segregation System

Khalifa University, 2026

Developed a dual-backbone model and functional prototype for four-category waste classification; presented at the World Future Energy Summit 2026.

Education

Universiti Sains Malaysia

Oct 2020 – Oct 2024

Bachelor of Computer Science (Honours), Intelligent Computing (AI) | Minor in Mathematics

Presentations and recognition: World Future Energy Summit 2026 (IntelliBin); UMEX & SimTEX 2026 (PRISM); Intern Excellence Award; Dean's Award List (four semesters); Gold Award.

Languages: English (IELTS Band 8, C1); Urdu and Hindi (native).